

Module/Pathway/Taxon Review:

alginate_polymerization_export in

Pseudomonas putida KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK, NCBI taxon 160488, proteome UP000000556) **Target bucket:** KEGG ppu00543 "Exopolysaccharide biosynthesis" (11 primary candidate genes) **Module reviewed:** alginate_polymerization_export **Review type:** Species-aware manual satisfiability and annotation-curation review · 2026-08-08

Summary

The **alginate_polymerization_export module is SATISFIED and functional** in *Pseudomonas putida* KT2440. The genome carries a single, contiguous alginate biosynthesis operon, **PP_1277–PP_1288**, that encodes one-to-one orthologs of every canonical polymerization and export step: the inner-membrane glycosyltransferase Alg8, the c-di-GMP-responsive copolymerase Alg44, the periplasmic scaffold proteins AlgK and AlgX, the mannuronan C5-epimerase AlgG, the alginate lyase AlgL, the O-acetylation machinery AlgI/AlgJ/AlgF, and the outer-membrane export porin AlgE. Precursor synthesis (AlgA, AlgD) and an AlgC-type phosphomannomutase are also present. The module can be marked **covered** with high confidence, with mechanism transferred from the essentially fully orthologous *P. aeruginosa* machinery.

The commissioned KEGG ppu00543 candidate list is, however, a poor proxy for the module and must be corrected. Of the 11 candidate genes, only 6 are genuine alginate genes. The bucket **omits four essential module genes** that are present in the genome (algL/PP_1281, algG/PP_1283, algE/PP_1284, algK/PP_1285) and **includes five over-propagated non-alginate candidates**: the four serine O-acetyltransferase / CysE proteins PP_0228, PP_0840 (cysE), PP_1110, PP_3136 (EC 2.3.1.30, cysteine biosynthesis, KO K00640), plus PP_2124 (KO K21001 = PslH, a distinct Psl-type EPS system). These over-propagations arise because KEGG map 00543 aggregates all EPS-related O-acetyl and glycosyltransferase activities, which sweeps in unrelated enzymes that share an EC number or a generic KO.

Functionally, the machinery is not merely a genomic relic. Direct experiments in *P. putida* show total EPS and alginate production increase specifically under **matric (water-limiting) stress**, that an *algD* mutant forms structurally altered biofilms with reduced desiccation survival, and that the single KT2440 *alg* cluster contributes to drought-protective biofilm formation on plants. The recommended curation action is: mark the module **covered**, **add** the four missing *alg* genes, **drop** the five over-propagated candidates, and promote the four newly added genes plus AlgX to full `fetch-gene` review.

Target-Organism Pathway Definition

What the pathway includes

The alginate_polymerization_export module covers the **membrane and periplasmic steps that convert the activated nucleotide-sugar precursor GDP-mannuronate into secreted, chemically modified alginate polymer**. In the synthase-dependent (Wzx/Wzy-independent) secretion paradigm used by *Pseudomonas*, this is a single envelope-spanning multiprotein complex that performs:

1. **Polymerization** at the inner membrane — Alg8 (GT2 glycosyltransferase) catalyzes β -1,4 linkage of mannuronate units, activated post-translationally by the copolymerase Alg44, whose PilZ domain binds c-di-GMP.
2. **Periplasmic guidance/scaffolding** — AlgK (TPR lipoprotein) and AlgX bridge the periplasm and pre-orient the nascent chain for export; AlgX additionally acts as an SGNH O-acetyltransferase.
3. **Epimerization** — AlgG, the mannuronan C5-epimerase, converts β -D-mannuronate (M) to α -L-guluronate (G), tuning polymer mechanics.
4. **Chain-length/quality control** — AlgL, a periplasmic alginate lyase, degrades mislocalized polymer and protects the periplasm.
5. **O-acetylation** — AlgI (integral membrane), AlgJ and AlgF add O-acetyl groups to mannuronate residues, protecting alginate from lyase degradation and modulating biofilm architecture.
6. **Outer-membrane export** — AlgE, an 18-stranded β -barrel porin, translocates the polyanionic chain across the outer membrane.

Neighboring processes to keep separate

- **Precursor biosynthesis (fructose-6-P → GDP-mannuronate):** AlgA (PP_1277, Man-6-P isomerase / Man-1-P guanylyltransferase), AlgC-type phosphomannomutase (PP_1777, PP_5288, KO K15778), and AlgD (PP_1288, GDP-mannose 6-dehydrogenase). AlgC is pleiotropic (shared with LPS biosynthesis) and lies outside the cluster.
- **Cysteine biosynthesis (serine → O-acetyl-serine):** the CysE serine O-acetyltransferases (EC 2.3.1.30). These are the over-propagation source.
- **Regulation:** AlgU/AlgT (σ 22), MucABCD, AlgR, and c-di-GMP turnover (e.g., PP_4959). Not part of the structural module.
- **Other KT2440 EPS systems:** Psl (PP_2124/PslH), cellulose (*bcs*), and the c-di-GMP-induced surface polysaccharide cluster PP3133–PP3141. KEGG map00543 lumps all EPS together; alginate must be resolved out.

Alternate names / database definitions

KEGG defines the bucket as **map00543 "Exopolysaccharide biosynthesis"** — a broad aggregate map, NOT an alginate-specific module; there is no dedicated alginate KEGG module, which is the root cause of the mismatch. The alginate genes follow the canonical *P. aeruginosa* operon naming **algD-8-44-K-E-G-X-L-I-J-F-A**; in KT2440 the operon is the **reverse-strand mirror** running PP_1277 (algA) → PP_1288 (algD). UniProt uses "Alginate biosynthesis protein" descriptors for most members.

Expected Step Model

Step	Function	Enzyme	KT2440 locus	UniProt	KO	In bucket?
Precursor	Man-6-P isomerase / Man-1-P guanylyltransferase	AlgA	PP_1277	—	K16011	No (flanking)
O-acetylation	Alginate O-acetyl accessory	AlgF	PP_1278	Q88ND4	K19296	Yes
O-acetylation	Alginate O-acetylase (SGNH)	AlgJ	PP_1279	Q88ND3	K19295	Yes
O-acetylation	Alginate O-acetylase (membrane)	AlgI	PP_1280	Q88ND2	K19294	Yes
Quality control	Alginate lyase	AlgL	PP_1281	Q88ND1	K01729	Missing
Scaffold / O-acetyl	Periplasmic O-acetyltransferase (SGNH)	AlgX	PP_1282	Q88ND0	K19293	Yes
Epimerization	Mannuronan C5-epimerase	AlgG	PP_1283	Q88NC9	K01795	Missing
Export	Outer-membrane β -barrel porin	AlgE	PP_1284	Q88NC8	K16081	Missing
Scaffold	Periplasmic TPR lipoprotein	AlgK	PP_1285	Q88NC7	K19292	Missing
Polymerization	c-di-GMP copolymerase (PilZ)	Alg44	PP_1286	Q88NC6	K19291	Yes
Polymerization	GT2 glycosyltransferase synthase	Alg8	PP_1287	Q88NC5	K19290	Yes
Precursor	GDP-mannose 6-dehydrogenase	AlgD	PP_1288	—	K00066	No (flanking)
Precursor (ectopic)	Phosphomannomutase (AlgC-type)	AlgC	PP_1777, PP_5288	—	K15778	No

Verdict: Every expected polymerization/export/modification step has a dedicated, contiguous ortholog. The module is genomically complete.

Key Findings

Finding 1 — The complete canonical alginate operon is present and contiguous (PP_1277–PP_1288)

KEGG/GenBank genome annotation of KT2440 shows a single contiguous *alg* cluster, **PP_1277–PP_1288**, encoding all 12 canonical alginate genes: PP_1277 *algA* (Man-6-P isomerase/Man-1-P guanylyltransferase, EC 5.3.1.8/2.7.7.13, K16011), PP_1278 *algF* (K19296), PP_1279 *algJ* (K19295), PP_1280 *algI* (K19294), PP_1281 *algL* (alginate lyase, EC 4.2.2.3, K01729), PP_1282 *algX* (K19293), PP_1283 *algG* (poly-ManA C5 epimerase, EC 5.1.3.37, K01795), PP_1284 *algE* (outer-membrane porin, K16081), PP_1285 *algK* (scaffold lipoprotein, K19292), PP_1286 *alg44* (copolymerase, K19291), PP_1287 *alg8* (GT2 polymerase, K19290), and PP_1288 *algD* (GDP-mannose 6-dehydrogenase, EC 1.1.1.132, K00066). Gene order is the reverse-strand mirror of the *P. aeruginosa* *algD-8-44-K-E-G-X-L-I-J-F-A* operon. Every module step therefore has a dedicated ortholog with no gaps.

This is corroborated by direct KT2440 studies. [PMID: 21507178](#) states: "*The gene clusters *alg* and *bcs*, which code for proteins mediating alginate and cellulose biosynthesis, were found to play minor roles in *P. putida* KT2440 biofilm formation and stability under the conditions tested*" — confirming KT2440 carries an intact *alg* cluster (its phenotypic role being condition-dependent). [PMID: 41554215](#) describes "*biofilm-deficient mutants KT2440 Alg, with only one gene cluster for the exopolysaccharide alginate deleted*" — confirming a single, deletable-as-a-unit alginate cluster consistent with the contiguous PP_1277–PP_1288 operon.

Finding 2 — The KEGG ppu00543 bucket omits 4 core module genes and includes 5 over-propagated candidates

The commissioned candidate list (KEGG map ppu00543) contains 11 genes but is a poor match to the module.

- **False positives / over-propagation:** PP_0228, PP_0840 (*cysE*), PP_1110 and PP_3136 are all serine O-acetyltransferase / *CysE* (EC 2.3.1.30, KO K00640) — cysteine biosynthesis enzymes

mapped onto the EPS O-acetyl node; none is an alginate gene. PP_2124 (KO K21001, generic glycosyltransferase / PslH) belongs to a different EPS system.

- **Missing from the bucket but present in the genome and central to the module:** PP_1281 *algL* (lyase, K01729), PP_1283 *algG* (C5 epimerase, K01795), PP_1284 *algE* (export porin, K16081), and PP_1285 *algK* (periplasmic scaffold, K19292).
- **Genuine alginate genes correctly in the bucket:** *algF* (PP_1278), *algJ* (PP_1279), *algI* (PP_1280), *algX* (PP_1282), *alg44* (PP_1286), *alg8* (PP_1287).

Four CysE paralogs collapsing onto a single EPS node is a textbook signature of EC/KO over-propagation on an aggregate overview map.

Finding 3 — UniProt independently confirms cluster functions; PP_2124 is PslH; AlgC-type PMM is ectopic

UniProt (reference proteome UP000000556) confirms each "missing-from-bucket" cluster gene: **Q88ND1 *algL*** = Alginate lyase (Lyase, Periplasm, Signal peptide); **Q88NC9 *algG*** = Mannuronan C5-epimerase (Alginate biosynthesis, Periplasm); **Q88NC8 *algE*** = Alginate production protein AlgE (Cell outer membrane); **Q88NC7 *algK*** = AlgK (Alginate biosynthesis, Lipoprotein, Palmitate, **3D-structure** cross-reference). Over-propagation candidates are confirmed non-alginate: **Q88L13 PP_2124** = "Glycosyl transferase" (keyword only "Transferase"; KO K21001 = *pslH*, mapping to map00543 EPS and map02025 biofilm — a different EPS system); **Q88PLO *cysE*** = serine O-acetyltransferase (Amino-acid biosynthesis, Cytoplasm — not alginate). The precursor phosphomannomutase (AlgC-type, bifunctional PMM/PGM, KO K15778) is encoded outside the *alg* cluster at PP_1777 and PP_5288.

Finding 4 — The machinery is functional and conditionally induced by matric/water-limiting stress

In *P. putida* (strain mt-2, the parent lineage of KT2440), total EPS and alginate production increase with matric (water-limiting) but not solute stress; an *algD* mutant forms structurally altered biofilms and shows reduced desiccation survival ([PMID: 17601783](#)). That study reports: "*Total exopolysaccharide (EPS) and alginate production increased with increasing matric, but not solute, stress severity, and alginate was a significant component, but not the major component, of EPS,*" and "*Alginate deficiency decreased survival of desiccation not only by *P. putida* but also by *Pseudomonas aeruginosa* PAO1 and *Pseudomonas syringae* pv. *syringae* B728a.*" A matric-stress genetic screen in *P. putida* independently recovered alginate-biosynthesis genes among

desiccation-tolerance determinants ([PMID: 15101980](#)). KT2440-specific work shows the single *alg* cluster contributes to biofilm stability (minor under standard lab conditions; [PMID: 21507178](#)) and to drought-protective biofilm on plants ([PMID: 41554215](#)).

Candidate Genes and Evidence

Genuine alginate genes correctly in the bucket (retain — high confidence)

- **alg8 / PP_1287 (Q88NC5)** — GT2 alginate polymerase catalytic subunit; membrane-embedded; directly interacts with Alg44 ([PMID: 25968647](#)). *Caveat*: broad EC 2.4.-.- placeholder — the true alginate synthase reaction lacks a clean EC.
- **alg44 / PP_1286 (Q88NC6)** — inner-membrane copolymerase with N-terminal PilZ domain; binds **dimeric c-di-GMP** to post-translationally activate polymerization ([PMID: 25817996](#)).
- **algX / PP_1282 (Q88ND0)** — periplasmic SGNH O-acetyltransferase + chain protection; part of the AlgK–AlgX–MucD complex ([PMID: 21713511](#)). *Caveat*: broad EC 2.3.1.- drives its bucket assignment.
- **algI / PP_1280 (Q88ND2)** — polytopic membrane O-acetylation component.
- **algJ / PP_1279 (Q88ND3)** — SGNH-family O-acetyltransferase.
- **algF / PP_1278 (Q88ND4)** — O-acetylation accessory/modulator.

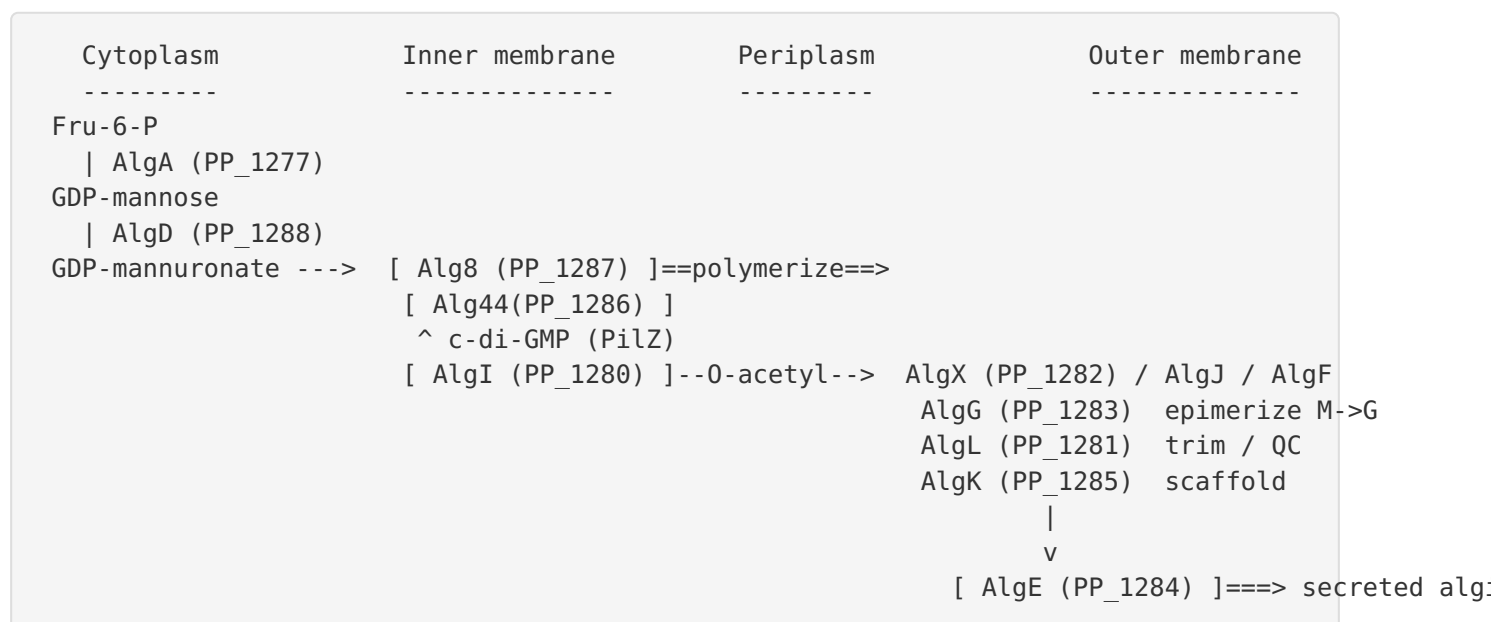
Core module genes MISSING from the bucket (add — high confidence)

- **algK / PP_1285 (Q88NC7)** — TPR lipoprotein scaffold; its loss destabilizes Alg8/Alg44 and blocks polymerization ([PMID: 23503314](#)).
- **algE / PP_1284 (Q88NC8)** — 18-stranded β -barrel export porin; structurally solved; complexes with AlgK ([PMID: 21778407](#), [PMID: 25084326](#), [PMID: 23335756](#)).
- **algG / PP_1283 (Q88NC9)** — poly-mannuronate C5-epimerase (EC 5.1.3.37); periplasmic chain modification.
- **algL / PP_1281 (Q88ND1)** — alginate lyase (EC 4.2.2.3); periplasmic quality control.

Over-propagated / wrong-system candidates (exclude)

Locus	UniProt	Annotation	Actual system	KO	Why over-propagated
PP_0228	Q88RA5	serine O-acetyltransferase (EC 2.3.1.30)	Cysteine biosynthesis	K00640	EC 2.3.1.30 → EPS O-acetyl node
PP_0840 (cysE)	Q88PL0	serine O-acetyltransferase (EC 2.3.1.30)	Cysteine biosynthesis (cytoplasm)	K00640	Canonical CysE; not alginate
PP_1110	Q88NU4	serine O-acetyltransferase (EC 2.3.1.30)	Cysteine biosynthesis	K00640	EC-based mis-map
PP_3136	Q88I65	Serine acetyltransferase (EC 2.3.1.30)	Cysteine biosynthesis	K00640	EC-based mis-map
PP_2124	Q88L13	Glycosyl transferase (generic)	Psl EPS system	K21001 (pslH)	Generic GT KO maps to map00543 + map02025

Mechanistic Model / Interpretation



The KT2440 machinery is the reverse-strand mirror of the *P. aeruginosa* algD-8-44-K-E-G-X-L-I-J-F-A operon. Because every subunit has a one-to-one *P. aeruginosa* ortholog, the extensive mechanistic work on the *P. aeruginosa* envelope-spanning complex — AlgE–AlgK–AlgX–Alg44–Alg8–AlgG interactions (PMID: 23503314), c-di-GMP activation of Alg44 (PMID: 25817996), AlgE porin gating (PMID: 21778407), and polymerization/modification coupling (PMID: 25968647) — transfers with **high confidence** to KT2440 at the structural/mechanistic level. Transfer of the *phenotype* (when/whether alginate is made) is more nuanced: KT2440 makes alginate **conditionally**, chiefly under water-limiting stress, rather than the constitutive mucoidy of CF-adapted *P. aeruginosa*. Presence $\neq$ constitutive activity; the module should be annotated as intact-but-regulated, not as a gap.

Gaps, Ambiguities, and Likely Over-Annotations

- 1. Bucket-vs-module mismatch (primary issue).** KEGG map00543 is a pan-EPS overview; it neither delimits alginate cleanly nor captures the AlgK/AlgE/AlgG/AlgL steps. Relying on the bucket alone would drop 4 essential steps and inject 5 non-alginate genes (6/11 precision).
- 2. Broad EC/KO mappings.** algX (EC 2.3.1.-) and alg8 (EC 2.4.-) carry incomplete ECs; EC 2.3.1.30 (CysE) is the over-propagation source; K21001 (generic GT) pulls in PslH.

3. **Precursor boundary.** algA (PP_1277) and algD (PP_1288) flank the operon but belong to a precursor-supply module; algC (PP_1777/PP_5288) is genomically separate and pleiotropic. Whether the module includes precursor supply is a definitional choice, not a satisfiability gap.
4. **Condition-dependent expression (not a genomic gap).** The cluster is minor under standard lab conditions (PMID: 21507178) but required for drought-protective biofilm (PMID: 41554215) and c-di-GMP-inducible (PMID: 21554519).
5. **No paralog ambiguity among true alg genes.** Each step = one dedicated cluster gene; ambiguity is confined to the CysE/GT over-propagations, which are unrelated.

Module and GO-Curation Recommendations

Module element / gene	Recommended status
alg8 (PP_1287), alg44 (PP_1286), algX (PP_1282), algI/J/F (PP_1280/1279/1278)	covered
algK (PP_1285), algE (PP_1284), algG (PP_1283), algL (PP_1281)	covered — add to module (bucket omission corrected)
algA (PP_1277), algD (PP_1288) precursor supply	covered but out-of-scope (assign to precursor module)
CysE genes PP_0228/PP_0840/PP_1110/PP_3136; PP_2124 (PslH)	not_in_module (over-propagation; remove)

Concrete curation actions: 1. Mark the alginate_polymerization_export module **covered** for PSEPK. 2. **Add** algL (PP_1281), algG (PP_1283), algE (PP_1284), algK (PP_1285) to the module. 3. **Drop** PP_0228, PP_0840, PP_1110, PP_3136, PP_2124 (reassign CysE to cysteine biosynthesis; PP_2124 to a Psl/EPS module). 4. **module_needs_revision on the generic boundary:** create a dedicated alginate module document keyed to the **PP_1277–PP_1288 locus** rather than to KEGG ppu00543. 5. **GO curation:** ensure the added genes carry alginate-specific terms — GO:0042121 (alginic acid biosynthetic process), outer-membrane/export component for AlgE (GO:0009279), periplasm for AlgG/AlgL/AlgK, and poly- β -D-mannuronate C5-epimerase activity for AlgG. Do **not** propagate GO:0006535 (cysteine biosynthesis) into this module for the CysE genes. No new GO term requests appear necessary — existing alginate terms cover the module.

Genes to Promote to Full [fetch-gene](#) Review

1. **PP_1284 algE** — export porin; missing from bucket, functionally essential, structurally defined in *P. aeruginosa*. Confirm β -barrel/OMP annotation and signal peptide.
 2. **PP_1285 algK** — scaffold lipoprotein; confirm lipobox/TPR and periplasmic localization (3D structure available — strong evidence).
 3. **PP_1283 algG** — C5-epimerase; confirm EC 5.1.3.37 and catalytic residues.
 4. **PP_1281 algL** — alginate lyase; confirm EC 4.2.2.3 (avoid confusion with degradation-only lyases).
 5. **PP_1282 algX** — resolve broad EC 2.3.1.- and dual scaffold/O-acetyltransferase annotation.
 6. (Lower priority) **PP_2124** — reassign to its correct Psl/non-alginate EPS system; one representative CysE (e.g., PP_0840) documented as over-propagation to prevent re-inclusion.
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Evidence Base

Direct target-species evidence (*P. putida*)

PMID	Finding	Strength for KT2440
17601783	Alginate produced under matrix (not solute) stress; <i>algD</i> mutant → altered biofilm, reduced desiccation survival	Strong — <i>P. putida</i> mt-2 (KT2440 parent lineage)
15101980	Matrix-stress screen recovered alginate genes among desiccation determinants	Strong — <i>P. putida</i> , KT2440 genome homology
21507178	KT2440 carries an intact <i>alg</i> cluster; minor biofilm role under standard lab conditions	Direct — KT2440
41554215	Single deletable <i>alg</i> cluster; drought-protective biofilm on tomato	Direct — KT2440
21554519	High c-di-GMP overproduces a <i>different</i> KT2440 EPS (PP3133–3141), not alginate/cellulose	Direct — KT2440; supports keeping non-alginate EPS out of the module

Mechanistic transfer evidence (*P. aeruginosa*, homology-based, high-confidence transfer)

PMID	Contribution
23503314	Envelope-spanning multiprotein complex; AlgK deletion destabilizes Alg8/Alg44
25968647	Alg8–Alg44 direct interaction; polymerization/modification coupling (AlgG, AlgX)
21778407	AlgE 18-strand β -barrel export porin structure
25084326	AlgE conformational landscape / gating; AlgK pre-oriens polymer
23335756	AlgE dual role in secretion and complex stability
25817996	Dimeric c-di-GMP binding by Alg44 PilZ domain drives polymerization
21713511	Periplasmic AlgK–AlgX–MucD complex links stability to regulation

Species-transfer note: All mechanistic/structural detail derives from *P. aeruginosa* and transfers with **high confidence** because KT2440 possesses a complete, one-to-one orthologous operon. Direct KT2440 evidence confirms cluster presence and phenotypic relevance but not molecular mechanism; alginate chemical composition/acetylation in KT2440 has not been directly characterized and remains open.

Limitations and Knowledge Gaps

1. **No direct KT2440 biochemistry per gene.** Gene presence and operon structure are from genome annotation (KEGG/GenBank) and UniProt; individual KT2440 subunit functions are inferred from *P. aeruginosa* orthologs, not measured in KT2440.
 2. **M/G composition and acetylation of KT2440 alginate are uncharacterized.** AlgG epimerase and AlgI/J/F/AlgX acetylation activities are assumed from homology; the actual polymer chemistry in KT2440 is unknown.
 3. **Conditionality of expression.** Alginate is minor/absent under standard lab conditions and induced by matric stress; the precise regulatory triggers (AlgU/ σ 22, MucABCD, c-di-GMP pools) in KT2440 are not fully mapped.
 4. **Precursor-module boundary** (AlgA/AlgC/AlgD placement) is a definitional choice not resolved here.
 5. **PslH (PP_2124) reassignment** is inferred from KO K21001; a dedicated Psl-module review would confirm it.
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Proposed Follow-up Experiments / Actions

1. **Curation (immediate):** Apply the recommendations above — add algL/algG/algE/algK, drop the four CysE genes + PP_2124, mark module covered, flag KEGG map00543 as a non-specific bucket.
2. **fetch-gene promotions:** Run full gene review on PP_1281, PP_1283, PP_1284, PP_1285, PP_1282.
3. **Schema fix:** Add a rule that EC 2.3.1.30 (serine O-acetyltransferase/CysE) and generic glycosyltransferase KOs (e.g., K21001) should not seed alginate-module membership from KEGG map00543.

4. **Experimental (to resolve gaps):** Characterize KT2440 alginate M/G ratio and acetylation under matrix stress; construct single-gene deletions (Δ algE, Δ algK, Δ algG) in KT2440 to confirm export/scaffold/epimerase roles directly rather than by transfer.
 5. **Regulatory mapping:** Test AlgU/MucA and c-di-GMP dependence of the KT2440 *alg* operon under water-limiting conditions.
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Key References

- Chang WS *et al.* (2007) *Alginate production by Pseudomonas putida creates a hydrated microenvironment... under water-limiting conditions.* [PMID: 17601783](#) — direct target-species functional evidence.
- van de Mortel M & Halverson LJ (2004) *Cell envelope components contributing to biofilm growth and survival of Pseudomonas putida in low-water-content habitats.* [PMID: 15101980](#) — target-species.
- *Influence of putative exopolysaccharide genes on Pseudomonas putida KT2440 biofilm stability.* [PMID: 21507178](#) — direct KT2440.
- *Biofilm formation by Pseudomonas putida KT2440 contributes to tomato drought stress resilience.* [PMID: 41554215](#) — direct KT2440.
- Matilla MA *et al.* (2011) *Cyclic diguanylate turnover... in Pseudomonas putida.* [PMID: 21554519](#) — KT2440 regulation.
- Hay ID *et al.* (2013) *Insights into the assembly of the alginate biosynthesis machinery in Pseudomonas aeruginosa.* [PMID: 23503314](#)
- Moradali MF *et al.* (2015) *Alginate Polymerization and Modification Are Linked in Pseudomonas aeruginosa.* [PMID: 25968647](#)
- Whitney JC *et al.* (2011) *Structural basis for alginate secretion across the bacterial outer membrane.* [PMID: 21778407](#)
- Tan J *et al.* (2014) *A conformational landscape for alginate secretion across the outer membrane.* [PMID: 25084326](#)
- Rehman ZU & Rehm BHA (2013) *Dual roles of Pseudomonas aeruginosa AlgE.* [PMID: 23335756](#)
- Whitney JC *et al.* (2015) *Dimeric c-di-GMP is required for post-translational regulation of alginate production.* [PMID: 25817996](#)

- Hay ID *et al.* (2012) *Identification of a periplasmic AlgK-AlgX-MucD multiprotein complex.* PMID: 21713511
 - KEGG [ppu00543](#) and KT2440 GenBank locus annotations PP_1277–PP_1288; UniProt reference proteome UP000000556 (retrieved 2026-08-08).
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Prepared for manual module satisfiability and gene-annotation curation. Evidence tiers: KT2440-direct (genome/UniProt annotation + PMIDs 17601783, 15101980, 21507178, 41554215, 21554519) vs. P. aeruginosa mechanistic transfer (strong at structure/mechanism, not phenotype).

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